

Visual Running Biography: An Interactive Exploration of Personal Training History

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ABSTRACT

Fitness tracking platforms generate rich historical records of personal running data but present it only as activity feeds and aggregate statistics. This paper presents *Visual Running Biography*, an interactive web-based visualization that transforms a runner's Strava history into an explorable narrative organized around automatically detected training phases. The system applies the PELT changepoint detection algorithm to a multivariate weekly feature sequence to segment training history into labeled phases Base, Building, Peak, Sharpening and Recovery. These phases are presented through four coordinated views that support progressive exploration. It provides a phase overview timeline and a draggable zoom into any phase that allows users to examine how key metrics shift across transitions. It also includes a week-level statistics view and a day-level detail view with per-kilometer pace, heart rate, training zones, and running efficiency for individual runs. This design guides the user from high-level structure down to individual workout detail and enables runners to interpret their history as a coherent personal trajectory of development.

1 INTRODUCTION

Recreational runners accumulate detailed training histories on platforms like Strava, yet these tools surface data as a flat stream of individual activities. It remains difficult in practice understanding how training evolved as a whole, which metrics changed and how.

Running data naturally forms a narrative through cycles of build-up, peak and recovery. Making that structure visible requires algorithmic interpretation and interactive exploration. Personal data is increasingly understood as a medium for self-reflection alongside optimization [3]. In this work I treat training phases as the primary unit of meaning.

Visual Running Biography (VRB) [18] is an interactive web visualization that transforms a runner's Strava history into a narrative organized around automatically detected training phases. I apply changepoint detection to weekly training load to identify phases then present them through a four-level progressive drill-down from a phase overview down to individual run metrics. The design is grounded in Munzner's nested model [9] and draws on research in narrative [13] and sports [12] visualization.

This paper makes three contributions. First, a **design study** for a multi-level explorable narrative visualization of personal running training history. Second, an **automated phase detection pipeline** that labels training phases from weekly aggregated Strava data. Third, a **four-level progressive drill-down interaction model** moving from phase overview to per-kilometer run detail. The full system demo is available online [17].

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2 BACKGROUND AND RELATED WORK

2.1 Personal Informatics and Self-Tracking Visualization

Moore et al. [8] identified a gap in personal informatics and showed that existing tools focus on automated insight generation however do not support the open-ended exploration users actually want. Strömél et al. [15] showed that narrative representations of fitness data increase user engagement and reflection compared to standard charts alone. Together, these findings motivate the design of *VRB*.

2.2 Sports Data Visualization

Perin et al. [12] survey sports data visualization and note that most existing work targets team and match-level analysis and have less attention to individual athletes over longitudinal training periods. Li et al. [7] confirm this for running specifically across 47 papers and the predominant focus is on in-situ visualization during the run itself with little work on post-hoc reflection on training history. This gap is further acknowledged by Elvitigala et al. [5] and he identified that longitudinal performance tracking as one of the grand challenges in SportsHCI and he also noted that interactive systems rarely address how athletes develop and change over months and years. *VRB* directly targets this gap by organizing a runner's full history into an explorable phase-structured narrative.

2.3 Narrative Visualization

Segel and Heer [13] established the foundational taxonomy of narrative visualization. They distinguished author-driven from reader-driven structures as well as described a “martini glass” pattern where a constrained authored opening gradually broadens into free exploration. *VRB* follows the structure where the phase overview is fixed and authored and while the drill-down levels give the user increasing freedom to examine what feels personally meaningful.

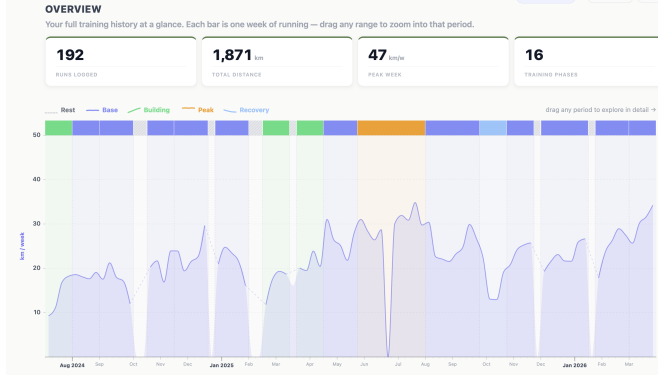
Bach et al. [1] identified design patterns for data-driven storytelling, visual transitions between scenes and progressive pacing. Oesch et al. [11] showed how scroll-based navigation can structure a narrative without requiring explicit choices at each step. My system borrows from this tradition and replaces scrolling with clicking as the navigation mechanism through the phase structure.

2.4 Progressive Disclosure and Hierarchical Interaction

Presenting everything at once overwhelms the user. Progressive disclosure addresses this by revealing information gradually through overview-plus-detail and drill-down patterns [4]. Shneiderman's visual information seeking mantra is overview first, zoom and filter, then details-on-demand [14] which captures the same principle that users with different levels of interest stop at different depths. *VRB* applies this structure across four levels of personal training data.

2.5 Automated Segmentation of Temporal Data

To move beyond manual annotation of training phases I applied changepoint detection method that identifies points in a time series where the underlying statistical properties shift. Truong et al. [20] review offline changepoint methods (including the PELT algorithm used here) which solves the multiple changepoint problem using a



(a) Training history overview.



(b) Phase timeline: distance and pace tabs.

Figure 1: Level 1 and 2 views of the drill-down hierarchy.

penalty term to control sensitivity. Teune et al. [16] applied change-point analysis to team sport training data and showed that it can identify meaningful transitions that aggregate statistics conceal.

3 DESIGN

3.1 Data Abstraction

Following Munzner’s framework [10] I translated personal running data into formal abstractions. The **activity table** is the raw input and contains one item per individual run with attributes such as distance, pace, heart rate and elevation gain. From this I aggregate weekly summaries into the **weekly table** which includes total distance, run count, average pace, heart rate and a computed efficiency score. The weekly table then serves as input to the phase detection pipeline producing the **phase table**. In this table each item represents a contiguous segment of weeks labeled with a phase type (Base, Building, Peak, Sharpening, Recovery or Inactive). The phase labels are derived from established endurance training periodization concepts. Base, Building, Peak and Recovery reflect common training phases described in the literature [2]. *Sharpening* and *Inactive* are operational labels introduced for visualization purposes. So all three datasets form a natural hierarchy so that phases contain weeks, weeks contain activities.

Key attributes are quantitative (distance, pace, heart rate, elevation), categorical (phase type, activity classification) and temporal (week, date). Heart rate zones and activity classifications are not using fixed global standards and are computed relative to each user’s own history.

3.2 Task and Intent Abstraction

The primary user intent is **retrospective reflection** to understand how training evolved over time. Following Munzner [10] the main abstract tasks are to *discover trends* in temporal attributes, *identify features* such as phases and changepoints, *compare attributes across phases*, *browse* the data hierarchically without a predefined query and *explore* run details. The task space is open-ended and personally contextualized so all interpretations are relative to the user’s own history.

3.3 Modality

The system is a desktop-oriented web visualization with multiple coordinated views. Activity data is fetched from the Strava

API, preprocessed into structured JSON and loaded client-side for latency-free exploration.

The system targets recreational runners with accumulated training history both beginners who are curious about their progress and experienced runners who want to analyze specific metrics, plan future training cycles or share their development with others. Also the system targets coaches who are reviewing an athlete’s development. Both can reflect on how their training evolved over time, notice changes in volume, pace and intensity. They can also see some transitions between distinct training phases. A typical session starts from the phase-level overview and progressively narrows to weeks and individual runs to contextualize observed patterns.

3.4 Visual Encoding Design

Each level of the drill-down uses different visual encodings depending on what the user needs to see at that scale. In general, I use **position** to encode quantitative values and **color** to distinguish categories [6].

Level 1: Training History Overview. The overview encodes the full training history as a **line and area chart** with time on the horizontal axis and weekly distance on the vertical axis (Figure 1a). A bar chart was rejected because at the scale of 80-100 weeks individual bars become too narrow and the continuous line better communicates temporal flow. Training phases are encoded as **contiguous colored bands** spatially aligned with the time axis so each phase can be read directly against the volume curve. Color hue encodes phase type categorically (Building: green, Peak: orange, Recovery: light blue, Base: purple, Rest: grey) and follows the **layered annotation strategy** of Segel and Heer [13]. Four summary statistics are displayed as **large-type cards** above the chart to provide quantitative anchors before exploration.

Level 2: Phase Timeline. When the user selects a time range the view switches to a phase timeline at week granularity. Each phase is displayed as a labeled header with aggregate statistics per each week (duration, mean weekly distance, mean pace) (Figure 1b). Weekly volume is encoded as a **stacked bar chart** and here the stacked layout encodes **run intensity** (easy: blue, moderate: yellow, hard: pink) and it makes effort distribution visible within each week. The timeline supports toggling between Distance, Pace, Avg HR and Efficiency. Each metric is shown as a **line chart** overlaid on the phase structure and it makes trends easier to see in continuous weekly data. Encoding trends as connected position along a

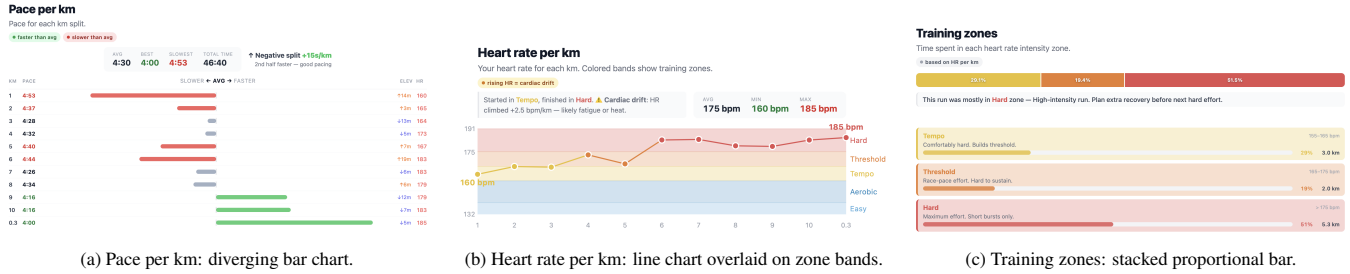


Figure 2: Run detail sub-views at Level 4 - Run Details.

common scale is the most accurate channel for detecting slope and direction [6]. The **pace axis is inverted** (faster at top) so that upward trends correspond intuitively to improvement.

Percentage-change badges at each phase boundary encode the direction and magnitude of change in key metrics using color-coded signed values (green for increase, red for decrease) and it makes cross-phase comparison a preattentive task. Initially these values were hidden behind a hover interaction but this proved too easy to miss and most users did not discover them so they were moved to the main view.

Level 3: Week Detail. The week detail view presents a **calendar grid** with one column per day and one card per run. Rest days appear as empty columns. Each card shows distance, pace, duration and elevation and is annotated with a **run type badge** and an **HR zone badge** using both color and text description.

Level 4: Run Detail. The run detail view exposes four sub-views via tabs which are Pace, Zones, Heart Rate and Efficiency. **Relative values are prioritized over absolute ones** across all sub-views and reflect how runners naturally evaluate a workout by comparing segments.

Pace per km uses a **diverging bar chart** centered on the run’s average pace, where bars extending left (red) for slower splits and right (green) for faster (Figure 2a). This emphasizes deviation from the mean and is more meaningful for identifying pacing strategies such as negative splits or fatigue than a standard line chart.

Training zones are encoded as a **stacked proportional bar** showing time spent in each heart rate zone (Figure 2c). This suits the compositional task of understanding effort distribution.

Heart rate per km is a **line chart** overlaid on colored zone bands (Figure 2b). It is appropriate for a continuous signal as the task is to observe trends and zone crossings. The background bands remove the need to reference a legend separately.

Efficiency per km reuses the **diverging bar layout** from pace and is centered on the mean efficiency index (speed divided by heart rate). Since efficiency has no intuitive absolute scale, deviation from the run mean is more readable than raw values. A line chart was rejected because it emphasizes trajectory instead of per-kilometer deviation from a reference point.

Across all levels textual labels and plain-language descriptions accompany the visual encodings so that the interface reads as a whole explanation.

3.5 Interaction Design

Interaction is based on **direct manipulation**. Users drag a range on the overview to select a period, click a week bar to open the week detail and click a run card to open the run detail. Each action narrows the scope without replacing the higher-level view so the navigation is constrained and reversible

Tooltips serve as a **secondary information layer**. By hovering over phase bands, intensity categories and HR zones it displays plain-language descriptions and by hovering over data points it reveals additional metrics at each level [4]. Plain-language labels are

also present on the main screen, for example, phase cards describe what each phase means in plain terms so that even users unfamiliar with training terminology can interpret the data without prior domain knowledge. The metric toggle at Level 2 and tabs at Level 4 switch the displayed data without leaving the current view so the surrounding context stays fixed.

The system includes a **guided tour** triggered on first use that introduces key interaction patterns before exploration begins.

3.6 Animation Design

Animation is used mostly to support continuity between views rather than to introduce additional information. Transitions between levels are implemented as **smooth scroll animations** and it signals a change in representation. Subtle animation effects are applied to interactive elements throughout the interface to improve visual polish, such as hover lifts on phase cards and week calendar entries. Also the guided tour uses **step-by-step animated high-lights** to draw attention to interactive elements in sequence. The design avoids complex or decorative animation to ensure that motion remains functionally aligned with interaction.

4 IMPLEMENTATION

The system is implemented as a **two-stage pipeline** that separates data processing from visualization. [19]

Backend. The backend is implemented in Python and combines data retrieval, processing and segmentation into a deterministic pipeline. Activity data is obtained through the Strava API via an OAuth flow. The pipeline uses pandas for aggregation and time-series manipulation, numpy for numerical operations and ruptures for changepoint detection. A Flask server handles API interaction and data processing requests.

Activities are first aggregated by ISO week. For each week the system computes total distance, run count, average pace, average run distance, long-run ratio and efficiency (speed relative to heart rate). A short-term trend feature is computed as the slope of weekly distance over the last four active weeks. Long inactivity gaps (≥ 10 days) are also detected and the data is split into independent segments.

Changepoint detection is applied to active weeks using the **PELT** (Pruned Exact Linear Time) algorithm [20] on a multivariate feature sequence which consists of total weekly distance, run count, average pace, long-run ratio, efficiency index and a short-term trend slope. PELT finds the optimal set of breakpoints by minimizing a cost function over all possible segmentations. PELT was chosen because its penalty term allows explicit control over segmentation sensitivity and prevents overly fine splits and also it is well suited to weekly aggregated training data with the moderate length of time series. Features are normalized using robust scaling Each feature centered by its median and divided by its interquartile range (75th minus 25th percentile). Each resulting segment is then labeled using rule-based classification based on relative volume and trend direction, assigning phase types (Base, Building, Peak, Sharpening,

Recovery or Inactive). Also short segments are merged to reduce noise. The output is a hierarchical structure (phases → weeks → activities) serialized as JSON.

Frontend. The frontend is implemented as a modular JavaScript application using D3.js for visualization rendering. It operates on precomputed JSON data and maintains a centralized application state that controls the current selection. Views are implemented as independent components that read from this state and update when it changes. Rendering is performed using SVG with scales and layouts computed dynamically based on the selected data subset. Updates are triggered by event-driven state changes.

User authentication and data loading are integrated with the Strava API so it allows users to connect their accounts and load their personal activity history directly into the system.

5 PROPOSED EVALUATION

The system is planned to be evaluated through a qualitative user study designed to assess interpretability, usability, perceived usefulness and the validity of the automatically detected training phases.

Participants. The study targets recreational runners who regularly track their activity using Strava and have accumulated at least 4 months of training history. A sample of 5–7 participants is appropriate for this exploratory evaluation.

Procedure. Each participant connects their own Strava account and explores the visualization in a think-aloud setting with no prior instructions beyond the built-in guided tour. Sessions are expected to last approximately 30–45 minutes. After free exploration participants answer structured questions covering five areas: (1) **phase validity** - whether detected labels match their recollection of each period, (2) **new observations** - patterns not previously noticed in standard Strava views, (3) **usability** - navigation clarity at each level using a 5-point Likert scale, (4) **perceived usefulness** - whether they would use the system for future training reflection and (5) **value assessment** - which views or features participants found most informative and which felt redundant or unclear. The last area is intended to identify which parts of the system add meaningful value and which may need to be redesigned or removed. Think-aloud recordings and post-session responses are analyzed qualitatively to identify usability issues and mismatches between detected and perceived phases.

6 DISCUSSION & FUTURE WORK

6.1 Lessons Learned

Several design decisions evolved significantly during development. One example is the percentage-change badges at phase boundaries which were initially hidden behind a hover interaction but proved too easy to miss and were later moved to the main view. This reinforced a broader lesson that information which is central to the narrative should not require discovery.

A second insight that emerged during implementation is that relative values are more interpretable than absolute ones. It was more effective to encode pace and efficiency as deviation from a personal average instead of raw numbers as this made the charts immediately readable without requiring domain knowledge.

Finally, designing four levels of detail forced a constant trade-off between showing enough context and avoiding information overload. The solution was to commit fully to progressive disclosure making each level shows only what is relevant at that scale, and try not to repeat information that was shown before.

6.2 Broader Design Implications

This work suggests that personal longitudinal data becomes easier to understand when organized into meaningful intermediate structures and not isolated events. In this system, training phases connect individual runs to larger patterns over time and this principle that may transfer to other domains involving personal data.

Also the design demonstrates the value of balancing accessibility and depth. High-level summaries support less experienced users, while detailed views remain available for more advanced ones and allows the same system to serve different levels of expertise.

Finally, the project highlights the importance of balancing narrative and statistics. Phase labels and textual descriptions give the data a storyline, at the same time when the underlying charts remain available for users who want to verify or explore further on their own.

6.3 Limitations

The system has several limitations. First, the phase detection approach is fixed and does not account for user-specific priorities. The segmentation is based on a predefined combination of features but in practice users may consider different aspects of training as more important. As a result, the detected phases reflect a single interpretation of the data.

Second, the system does not support direct comparison between arbitrary time periods or phases. Users are limited to sequential exploration and it makes it difficult to explicitly compare different segments of their training history.

Third, the quality of the analysis depends on the availability of heart rate data. In its absence the analysis relies on a reduced set of features and it may limit the accuracy.

Finally, the system has so far been tested only on running data. The underlying pipeline is not specific to running but its current feature set and phase labels are designed for running training structure. As a result it may not generalize to other endurance sports without further adaptation.

6.4 Future Work

Future work will focus on making the phase detection more flexible and user-driven in order to allow users to prioritize different aspects of their training, for example, weighting pace more heavily than distance or vice versa, and generate alternative phase segmentations accordingly.

A natural extension is to support explicit comparison between arbitrary time periods and let users select and overlay different phases or intervals to analyze how training has changed.

A particularly interesting extension would be to integrate an LLM-based analysis layer that takes the detected phase structure as context and generates personalized narrative summaries of the athlete's training history. This could help users who are less comfortable reading charts to still extract meaningful insights from their data.

Finally, the approach could be extended to additional data sources beyond Strava, such as Garmin or Apple Health to allow users to build a complete training history regardless of platform.

7 CONCLUSION

Visual Running Biography presents a design and implementation of an interactive visualization system that transforms longitudinal running data into an explorable, phase-structured narrative. By applying changepoint detection to weekly training load and presenting the detected structure through four coordinated, progressive levels of detail, the system enables runners to interpret their training history not as isolated activities but as a coherent personal trajectory. The design is grounded in Munzner's nested model and draws on principles from narrative visualization, progressive disclosure and personal informatics. A proposed qualitative evaluation with recreational runners will assess the validity of the detected phase structure and the overall usability of the drill-down interaction model.

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